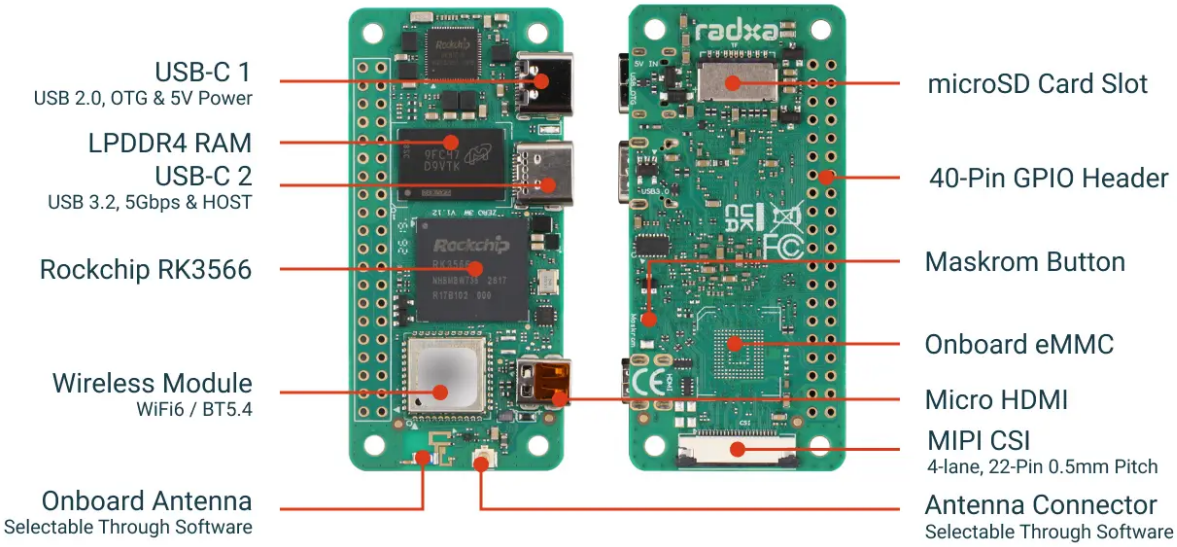
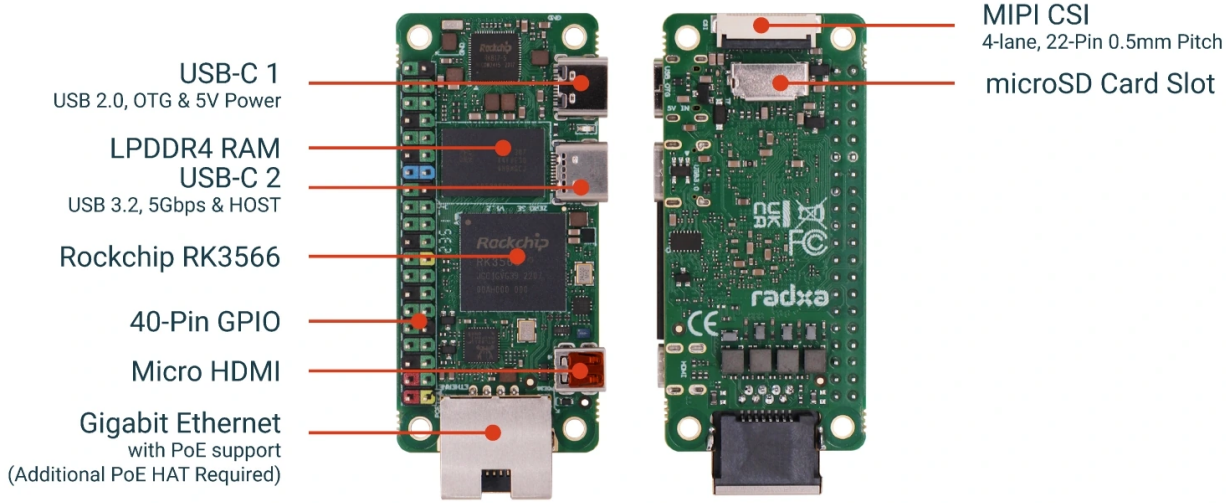
Hardware interface description

1. Interface overview

- **Radxa ZERO 3W**



- **Radxa ZERO 3E**



2. Power

The ZERO 3W / 3E is powered via the **USB-C 1** port (USB 2.0 OTG) and only supports **5V** input. A 5V/2A or higher power adapter is recommended.

> **TIP**
>
> Radxa recommends Radxa Power PD30W.

3. Debug Serial Port

The debug serial port of ZERO 3 uses **UART2_M0** (40-pin GPIO Pin 8: TX, Pin 10: RX), with a default baud rate of 1500000n8.

> **UART2_M0 IS NOT RECOMMENDED FOR USE AS A NORMAL UART**
>
> UART2_M0 is also the default U-Boot console. If you enable UART2_M0 as a normal serial port through the rsetup overlay, input on the serial port will interrupt the boot process and the device may fail to boot.
>
> If you really need to use UART2_M0, first update the system and bootloader via `rsetup`, then edit `/etc/default/u-boot` to uncomment `prompt` and `timeout` and set them to 0, and finally run `sudo u-boot-update` to rebuild the boot configuration.

4. Ethernet Port

5. 40 PIN GPIO

5.1 GPIO voltage

GPIO	Voltage	Maximum
All GPIOs	3.3V	3.63V
SARADC_IN5	3.3V	3.3V

5.2 GPIO interface

Radxa ZERO 3 provides a 40-pin GPIO header, which is compatible with most SBC accessories on the market.

Tip: Actual compatibility depends on the use case.

> **TIP**
>
> Pin 3, Pin 5, Pin 27, and Pin 28 add extra pull-up resistors for I2C device power supply, so they work abnormally when used as GPIOs.

GPIO number	Function5	Function4	Function3	Function2	Function1	Pin#
					+3.3V	1
32				UART3_RX_M0	GPIO1_A0	3
33				UART3_TX_M0	GPIO1_A1	5
116		PWM14_M0			GPIO3_C4	7
					GND	9
97					GPIO3_A1	11
98		I2S3_MCLK_M0			GPIO3_A2	13
104					GPIO3_B0	15
					+3.3V	17
147		PWM15_IR_M1	I2S3_SCLK_M1	SPI3_MOSI_M1	GPIO4_C3	19
149	UART9_TX_M1	PWM12_M1	I2S3_SDO_M1	SPI3_MISO_M1	GPIO4_C5	21
146		PWM14_M1	I2S3_MCLK_M1	SPI3_CLK_M1	GPIO4_C2	23
					GND	25
138		I2C4_SDA_M0	I2S2_SDI_M1		GPIO4_B2	27

GPIO number	Function5	Function4	Function3	Function2	Function1	Pin#
107		I2C5_SCL_M0			GPIO3_B3	29
108		I2C5_SDA_M0			GPIO3_B4	31
115	UART5_RX_M1		I2S1_SCLK_RX_M2		GPIO3_C3	33
100			I2S3_LRCK_M0		GPIO3_A4	35
36			I2S1_SCLK_RX_M0		GPIO1_A4	37
					GND	39

5.3 USB Function Configuration on 40-Pin

On the ZERO 3W / 3E's 40-Pin interface, the following pins can be configured for USB 2.0:

- USB2_HOST2_DM: Pin number 28, resistor location R46.
- USB2_HOST2_DP: Pin number 27, resistor location R45.

By default, Pin-27 can be configured in software for functions such as GPIO4_B2 (see 40-Pin Pinout), while the USB2_HOST2_DP signal is not activated on the hardware. Pin-28 can be configured in software for functions such as GPIO4_B3 (see 40-Pin Pinout), and the USB2_HOST2_DM signal is also not activated on the hardware. To change these pins to USB functions, please follow the steps below to modify the reserved resistors:

- **Remove the 0-ohm resistors at R32 and R39.**
- **Solder 0-ohm resistors at R45 and R46.**

> **TIP**
>
> The schematic and resistor locations can be viewed and downloaded from the hardware documentation.Hardware Documentation Download

> **[OPERATION NOTE]**
>
> This procedure requires soldering skills, and it is recommended that it be performed by an experienced technician.

6. USB C

There are two USB Type-C ports: **USB-C 1** (USB 2.0 OTG, supports power and OTG) and **USB-C 2** (USB 3.0 HOST). The board is **powered via the USB-C 1 port**.

7. HDMI

Provides a Micro HDMI port.

8. MIPI CSI

The ZERO 3W / 3E provides one **4-lane MIPI CSI** interface with a **22-Pin, 0.5mm pitch** FPC connector.

PIN	Name
1	GND
2	MIPI_CSI_RX_D0N
3	MIPI_CSI_RX_D0P
4	GND
5	MIPI_CSI_RX_D1N
6	MIPI_CSI_RX_D1P
7	GND
8	MIPI_CSI_RX_CLK0N
9	MIPI_CSI_RX_CLK0P

PIN	Name
10	GND
11	MIPI_CSI_RX_D2N
12	MIPI_CSI_RX_D2P
13	GND
14	MIPI_CSI_RX_D3N
15	MIPI_CSI_RX_D3P
16	GND
17	CAMERAB_PDN_L
18	CIF_CLKOUT
19	GND
20	I2C2_SCL_M1
21	I2C2_SDA_M1
22	VCC_3V3

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